MCC Entrance Test, August 2026 — Day 2 — Solutions

Level T1

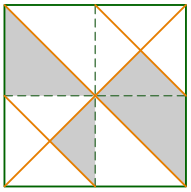
Test Information. Level T1 — 90 minutes.
6 problems: 2 Easy (10 pts), 2 Medium (15 pts, two parts), 2 Hard (25 pts, three parts); 100 points total.
Areas: Number Theory (×2), Geometry (×2), Algebra, Combinatorics.
Rules: show your work (partial credit rewards genuine progress); no calculators; read all six before choosing where to start; box your final answers.
A past MCC entrance test, free to try: https://nghia71.github.io/readiness

Problem 1 (10 pts).[1] A counting number is *marked* if it has an *odd* number of factors. (*For example, the number* 6 *has exactly four factors* – 1, 2, 3, *and* 6 – *so* 6 *is not marked.*) How many counting numbers less than 20 are marked?

Official solution. Factors of a number almost always come in pairs: if d is a factor of n, so is n/d, and normally $d \neq n/d$, so factors pair off and the total count is even. The only way a number can have an *odd* number of factors is if one of its factors pairs with *itself* – that is, $d = n/d$, which means $n = d^2$: a perfect square.

The counting numbers less than 20 that are perfect squares are $1, 4, 9, 16$ (since $1^2, 2^2, 3^2, 4^2$ are all less than 20, but $5^2 = 25$ is not). So $\boxed{4}$ numbers are marked. □

Problem 2 (10 pts).[2] A large square is divided into four smaller squares of equal size by its two midlines, and both full diagonals of the large square are drawn. In two of the small squares (diagonally opposite each other), only one diagonal passes through, splitting each into 2 triangles. In the other two small squares, a second diagonal is also drawn, splitting each of *those* into 4 triangles. In each small square, exactly one triangle is shaded, as shown. What fraction of the large square's area is shaded?

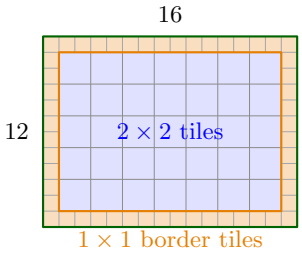


Official solution. Let the big square have side 2, so its area is 4 and each of the 4 small squares has area 1.

Two small squares have only *one* diagonal, so each splits into 2 triangles of area $\frac{1}{2}$ each; shading one triangle in each contributes $\frac{1}{2} + \frac{1}{2} = 1$.

The other two small squares have *both* diagonals, so each splits into 4 triangles of area $\frac{1}{4}$ each; shading one triangle in each contributes $\frac{1}{4} + \frac{1}{4} = \frac{1}{2}$.

Total shaded area $= 1 + \frac{1}{2} = \frac{3}{2}$, out of the big square's total area 4: $\frac{3/2}{4} = \boxed{\frac{3}{8}}$. □



Problem 3 (15 pts).[3] Tyler is tiling the floor of a 12-foot by 16-foot rectangular room. He plans to place 1-foot by 1-foot square tiles to form a border, one tile wide, running along all four edges of the room. He will then fill in the rest of the floor using 2-foot by 2-foot square tiles.

(a) What are the dimensions of the inner rectangle that will be covered with the 2-by-2 tiles? [7 pts]

(b) How many tiles in total (both the 1-by-1 border tiles and the 2-by-2 interior tiles together) will Tyler use to cover the whole floor? [8 pts]

Official solution. **Part (a).** The one-tile-wide border removes 1 ft from *each* side of *each* dimension – that is 2 ft total off the length and 2 ft total off the width, so inner rectangle $= (12-2)$ ft $\times (16-2)$ ft $= \boxed{10 \text{ ft} \times 14 \text{ ft}}$.

Part (b). Border tiles: the border's total area equals the whole floor's area minus the inner rectangle's area, $(12 \times 16) - (10 \times 14) = 192 - 140 = 52$ square feet, and each border tile covers 1 square foot, so there are 52 border tiles.

Interior tiles: the inner area is $10 \times 14 = 140$ square feet, and each 2×2 tile covers 4 square feet, so there are $140 \div 4 = 35$ interior tiles. Total tiles $= 52 + 35 = \boxed{87}$. □

[1] Math Olympiads for Elementary and Middle Schools (MOEMS). [2] UK Junior Mathematical Challenge. [3] American Mathematics Competition 8 (AMC 8).

Problem 4 (15 pts).[4] A vendor sells three kinds of tickets: Type A tickets for \$3 each, Type B tickets for \$1.50 each, and Type C tickets for \$0.50 each. One evening, the vendor sells exactly 20 tickets in total – at least one of each kind – and collects exactly \$20 in all.

(a) Show that if b is the number of Type A tickets sold and m is the number of Type B tickets sold, then $5b + 2m = 20$. [5 pts]

(b) Find how many Type A tickets the vendor sold, and how many dollars this brought in. [10 pts]

Official solution. **Part (a).** Let b, m, p be the number of Type A, Type B, and Type C tickets sold. Then

$$b + m + p = 20 \qquad \text{and} \qquad 3b + 1.5m + 0.5p = 20.$$

Multiplying the second equation by 2 gives $6b + 3m + p = 40$. Subtracting the first equation:

$$(6b + 3m + p) - (b + m + p) = 40 - 20 \implies \boxed{5b + 2m = 20}.$$

Part (b). Since $2m$ and 20 are both even, $5b$ must be even, which forces b itself to be even. Since $5b \le 20$, $b \in \{0, 2, 4\}$; since the vendor sold at least one Type A ticket, $b \in \{2, 4\}$.

If $b = 2$: $m = (20 - 10)/2 = 5$, so $p = 20 - 2 - 5 = 13$. All three counts are at least 1. ✓

If $b = 4$: $m = (20 - 20)/2 = 0$ – but at least one Type B ticket must be sold, so this case is invalid.

So $b = 2$ is the only possibility. Type A tickets brought in $3 \times 2 = \boxed{\$6}$. □

Problem 5 (25 pts).[5] A survey asks each of 425 people which of three activities – A, B, and C – they have tried. In total, 351 people tried activity A, 71 tried activity B, and 203 tried activity C. Of these people, 199 tried more than one activity, and 8 tried none of the three activities at all.

(a) How many of the 425 people tried at least one of the three activities? [5 pts]

(b) If a people tried exactly two activities and t people tried all three activities, write the equation relating a, t, and the 199 figure above. [3 pts]

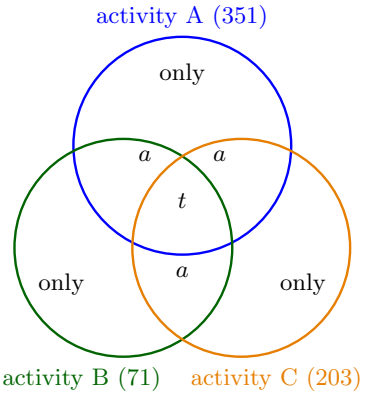
(c) Find t, the number of people who tried all three activities. [17 pts]

Official solution. **Part (a).** People who tried at least one activity = 425 total − 8 who tried none = $\boxed{417}$.

Part (b). "More than one activity" means exactly two, or exactly three: $\boxed{a + t = 199}$.

Part (c). By inclusion–exclusion, the count of "at least one activity" equals the sum of the three individual counts, minus the sum of the pairwise overlaps, plus the triple-overlap t (added back once, since it was subtracted three times by the pairwise term):

$$351 + 71 + 203 - (\text{sum of pairwise overlaps}) + t = 417$$
$$(\text{sum of pairwise overlaps}) - t = 625 - 417 = 208.$$



The key step: the "sum of pairwise overlaps" counts each exactly-two-activity person *once*, but counts each exactly-three-activity person *three times* (once per pair) – so sum of pairwise overlaps $= a + 3t$.

Substituting:

$$(a + 3t) - t = 208 \implies a + 2t = 208$$
$$a = 199 - t \implies (199 - t) + 2t = 208 \implies t = \boxed{9}.$$

□

[4] Hanoi Open Mathematics Competition. [5] Purple Comet! Math Meet, Middle School Division.

Problem 6 (25 pts).[6] Six contestants each compete in two rounds. Their round-1 scores are $23, 24, 26, 28, 28, 29$ – one score per contestant. Their round-2 scores are $10, 10, 12, 12, 14, 16$ – again one per contestant, but we are not told which round-2 score belongs to which contestant. The winner is whichever contestant has the highest combined score (round 1 + round 2).

(a) Add up all twelve scores to find the total of every contestant's combined score, added together. Use this total to explain why at least one contestant must have a combined score of 39 or more. [7 pts]

(b) Among the round-2 scores $(10, 10, 12, 12, 14, 16)$, how many are 11 or less? [2 pts]

(c) Show that a winning score of exactly 39 is impossible, and find the least possible combined score the winner could have – along with one matching of round-1 and round-2 scores that achieves it. [16 pts]

Official solution. **Part (a).** Sum of round-1 scores: $23 + 24 + 26 + 28 + 28 + 29 = 158$. Sum of round-2 scores: $10 + 10 + 12 + 12 + 14 + 16 = 74$. Grand total $= 158 + 74 = 232$.

If every contestant's combined score were 38 or less, the grand total could be at most $6 \times 38 = 228 < 232$ – a contradiction. So at least one combined score is $\boxed{\geq 39}$.

Part (b). Only $\boxed{2}$ round-2 scores are 11 or less: the two 10's. ($12, 12, 14, 16$ are all at least 12.)

Part (c). The three highest round-1 scores are $28, 28, 29$. Suppose every combined score were ≤ 39. Then the round-2 score paired with 29 must be ≤ 10, and the round-2 score paired with each 28 must be ≤ 11. So all *three* of these contestants would need a round-2 score from the set of values ≤ 11 – but by part (b), only *two* such values exist (the two 10's). Three contestants cannot be matched to only two available low scores – a contradiction. So a combined score of 39 for the winner is impossible, and the minimum possible winning score must be at least 40.

This bound is achievable: pairing $29 \leftrightarrow 10$ $(= 39)$, $28 \leftrightarrow 10$ $(= 38)$, $28 \leftrightarrow 12$ $(= 40)$, $26 \leftrightarrow 12$ $(= 38)$, $24 \leftrightarrow 14$ $(= 38)$, $23 \leftrightarrow 16$ $(= 39)$ uses every score exactly once and gives a maximum combined score of 40.

Least possible winning score $= \boxed{40}$. □

[6] MATHCOUNTS.